

# SAHAYAK — Phase 1 Setup Commands (Windows)

All commands verified and working as of June 2026

Run everything inside (sahayak\_env) unless stated otherwise

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## STEP 1 — Verify Python

```
python --version  
# Expected: Python 3.10.x or higher
```

## STEP 2 — Create project folder & virtual environment

```
mkdir sahayak  
cd sahayak  
python -m venv sahayak_env  
sahayak_env\Scripts\activate  
# You should now see (sahayak_env) at the start of your terminal
```

## STEP 3 — Upgrade pip

```
python -m pip install --upgrade pip
```

## STEP 4 — Create folder structure

```
mkdir utils notebooks data data\raw embeddings  
type nul > app.py  
type nul > .env  
type nul > utils\__init__.py  
type nul > utils\translator.py  
type nul > utils\search.py  
type nul > utils\llm.py  
type nul > utils\tts.py  
type nul > utils\stt.py
```

## STEP 5 — Install requirements.txt

```
pip install -r requirements.txt
```

⚠ NOTE: This does NOT fully work on Windows for all packages. Install the following manually as shown in steps below.

## STEP 6 — Install PyTorch (MUST use special URL on Windows)

```
pip install torch torchvision torchaudio --index-url
```

```
https://download.pytorch.org/whl/cpu
```

⚠ Do NOT use plain `pip install torch` on Windows — it will fail or install wrong version.

## STEP 7 — Install remaining packages that requirements.txt missed

```
pip install faiss-cpu
pip install openai-whisper
pip install gtts
pip install ollama
pip install transformers sentence-transformers accelerate
```

## STEP 8 — Install ffmpeg (run PowerShell as Administrator)

```
winget install ffmpeg
```

After install, close and reopen terminal, then verify:

```
ffmpeg -version
```

## STEP 9 — Clone IndicTrans2

```
git clone https://github.com/AI4Bharat/IndicTrans2
```

⚠ NOTE: The root of IndicTrans2 has NO setup.py or pyproject.toml. The huggingface\_interface subfolder is what we use — no pip install needed. We import models directly via HuggingFace transformers in the code.

## STEP 10 — Install Ollama

- Download from <https://ollama.com> and install like a normal Windows app
- Then open a NEW terminal and run:

```
ollama pull gemma2:2b
```

- Verify it downloaded:

```
ollama list
# Should show gemma2:2b in the list
```

## STEP 11 — Final verification (all must pass)

```
python -c "import streamlit, torch, faiss, transformers,
sentence_transformers, whisper, gtts, ollama; print('All core packages OK!)"
python -c "import torch; print('PyTorch:', torch.__version__)"
python -c "import faiss; print('FAISS OK!)"
python -c "import whisper; whisper.load_model('tiny'); print('Whisper OK!)"
```

```
python -c "from gtts import gTTS; print('gTTS OK')"  
streamlit hello
```

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### EVERY TIME YOU START A NEW SESSION — run this first:

```
cd C:\Users\knlwa\Desktop\GIT\SAHAYAK_AI\sahayak  
sahayak_env\Scripts\activate  
ollama serve    ← run this in a SEPARATE terminal window (keeps LLM running)
```

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### PACKAGES THAT WORKED FROM requirements.txt:

- streamlit
- pandas, numpy, tqdm
- beautifulsoup4, requests, lxml
- jupyter, ipykernel

### PACKAGES THAT NEEDED MANUAL INSTALL (Windows specific):

- torch → needs --index-url <https://download.pytorch.org/whl/cpu>
- faiss-cpu → not installed by requirements.txt
- openai-whisper → not installed by requirements.txt
- gTTS → not installed by requirements.txt
- ollama → not installed by requirements.txt
- transformers, sentence-transformers, accelerate → needed manual install